PYTHON FOR DATA SCIENCE · LESSON 6 OF 29

Lists: Creating, Slicing, Methods, Sorting & Comprehensions

Store and process ordered collections with lists. Index and slice them, add, remove and sort items, understand mutability, aliasing and copying (shallow vs deep), use nested lists, list comprehensions, and common patterns for cleaning and summarising data.

Python class 5 | Python 3.13 | Beginner | 120 min

By the end of this lesson you can

- Create lists and use indexing, slicing, concatenation, repetition and membership tests
- Apply list methods (append, extend, insert, remove, pop, sort, reverse, index, count) and built-ins (len, sum, min, max, sorted)
- Explain mutability, aliasing and the difference between assignment, shallow copy and deep copy
- Build lists with comprehensions, including conditions and nested loops
- Use lists for typical data tasks: filtering, ranking, de-duplicating while keeping order, flattening and unzipping

Datasets used

`employees_small.csv`

What is a list?

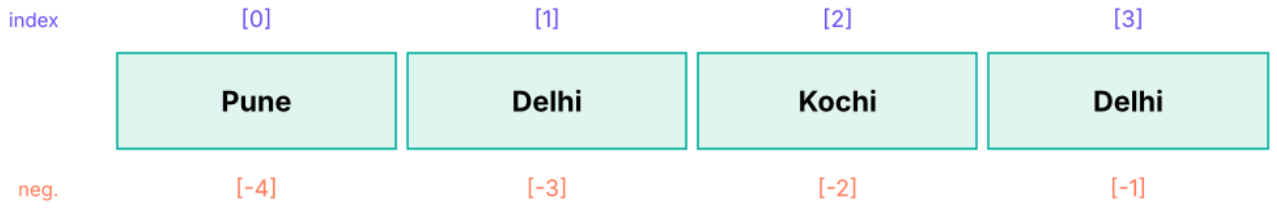
A **list** is an ordered, **mutable** collection of items written in square brackets. Items can be of any type, and may repeat.

Python

```
cities = ["Pune", "Delhi", "Kochi", "Delhi"]
mixed = [101, "Kavya", 97000.5, True, None]
empty = []
print(cities, len(cities), type(cities).__name__)
print(list("data"), list(range(5)), list((1, 2, 3)), [0] * 5)
```

OUTPUT

```
['Pune', 'Delhi', 'Kochi', 'Delhi'] 4 list
['d', 'a', 't', 'a'] [0, 1, 2, 3, 4] [1, 2, 3] [0, 0, 0, 0, 0]
```



A list is a row of numbered boxes. Items are referenced by position; the list itself can grow, shrink and change.

Indexing and slicing

Works exactly like strings (Lesson 3), but lists can also be **changed**.

Python

```python
nums = [10, 20, 30, 40, 50, 60]
print(nums[0], nums[-1], nums[1:4], nums[:2], nums[-2:], nums[::2], nums[::-1])
nums[0] = 99                    # change one item
nums[1:3] = [21, 31, 32]        # replace a slice (the length may change)
print(nums)
print(3 in nums, 99 in nums, nums.count(40), nums.index(40))
```

OUTPUT

```
10 60 [20, 30, 40] [10, 20] [50, 60] [10, 30, 50] [60, 50, 40, 30, 20, 10]
[99, 21, 31, 32, 40, 50, 60]
False True 1 4
```

Adding and removing items

Method	What it does	Example	Returns
`append(x)`	Add one item at the end	`a.append(5)`	None
`extend(iterable)` or `+=`	Add all items from another sequence	`a.extend([6, 7])`	None
`insert(i, x)`	Insert before position i	`a.insert(0, "first")`	None
`remove(x)`	Delete the first item equal to x	`a.remove(5)`	None (error if missing)
`pop(i=-1)`	Remove and return item at i	`last = a.pop()`	the item
`del a[i]` / `del a[i:j]`	Delete by position or slice	`del a[0]`	
`clear()`	Empty the list	`a.clear()`	None

Python

```python
a = [1, 2, 3]
a.append(4); a.extend([5, 6]); a.insert(0, 0); a += [7]
print(a)
print(a.pop(), a.pop(0), a)          # removes 7, then 0
a.remove(3); del a[0]; print(a)
print([1, 2] + [3], [1, 2] * 2)       # concatenation creates new lists
print(a.append(9))                    # None! methods that change a list return None
```

OUTPUT

```
[0, 1, 2, 3, 4, 5, 6, 7]
7 0 [1, 2, 3, 4, 5, 6]
[2, 4, 5, 6]
[1, 2, 3] [1, 2, 1, 2]
None
```

COMMON MISTAKES **`x = x.append(...)`**

`append`, `sort`, `reverse` and `extend` change the list **in place** and return `None`. Writing `nums = nums.sort()` leaves `nums` as `None`. Use `sorted(nums)` when you want a new list.

Sorting and reversing

Python

```
scores = [72, 95, 61, 88, 95]
print(sorted(scores), sorted(scores, reverse=True), scores)        # sorted() returns a
new list
scores.sort(); print(scores)                                        # .sort() changes the
list in place
words = ["banana", "Apple", "cherry", "apple"]
print(sorted(words), sorted(words, key=str.lower), sorted(words, key=len))
staff = [("Asha", 29, 61000), ("Ravi", 34, 88000), ("Meera", 41, 72000)]
print(sorted(staff, key=lambda p: p[2], reverse=True))              # sort records by the
3rd field
print(list(reversed([1, 2, 3])), min(staff, key=lambda p: p[1])[0])
```

OUTPUT

```
[61, 72, 88, 95, 95] [95, 95, 88, 72, 61] [72, 95, 61, 88, 95]
[61, 72, 88, 95, 95]
['Apple', 'apple', 'banana', 'cherry'] ['Apple', 'apple', 'banana', 'cherry']
['Apple', 'apple', 'banana', 'cherry']
[('Ravi', 34, 88000), ('Meera', 41, 72000), ('Asha', 29, 61000)]
[3, 2, 1] Asha
```

`sorted()` and `.sort()` are **stable**: equal items keep their original order, so you can sort by several keys by sorting repeatedly from the least to the most important key, or use a tuple key: `key=lambda p: (p[1], -p[2])`.

Built-in helpers

Python

```
data = [4, 8, 15, 16, 23, 42]
print(len(data), sum(data), min(data), max(data), sum(data) / len(data))
print(any(x > 40 for x in data), all(x > 0 for x in data))
print(list(enumerate(["a", "b"], 1)), list(zip([1, 2, 3], "abc")))
a, b, *rest = data; print(a, b, rest)                               # unpacking with *
first, *middle, last = data; print(first, middle, last)
```

OUTPUT

```
6 108 4 42 18.0
True True
[(1, 'a'), (2, 'b')] [(1, 'a'), (2, 'b'), (3, 'c')]
```

```
4 8 [15, 16, 23, 42]
4 [8, 15, 16, 23] 42
```

Mutability, aliasing and copying

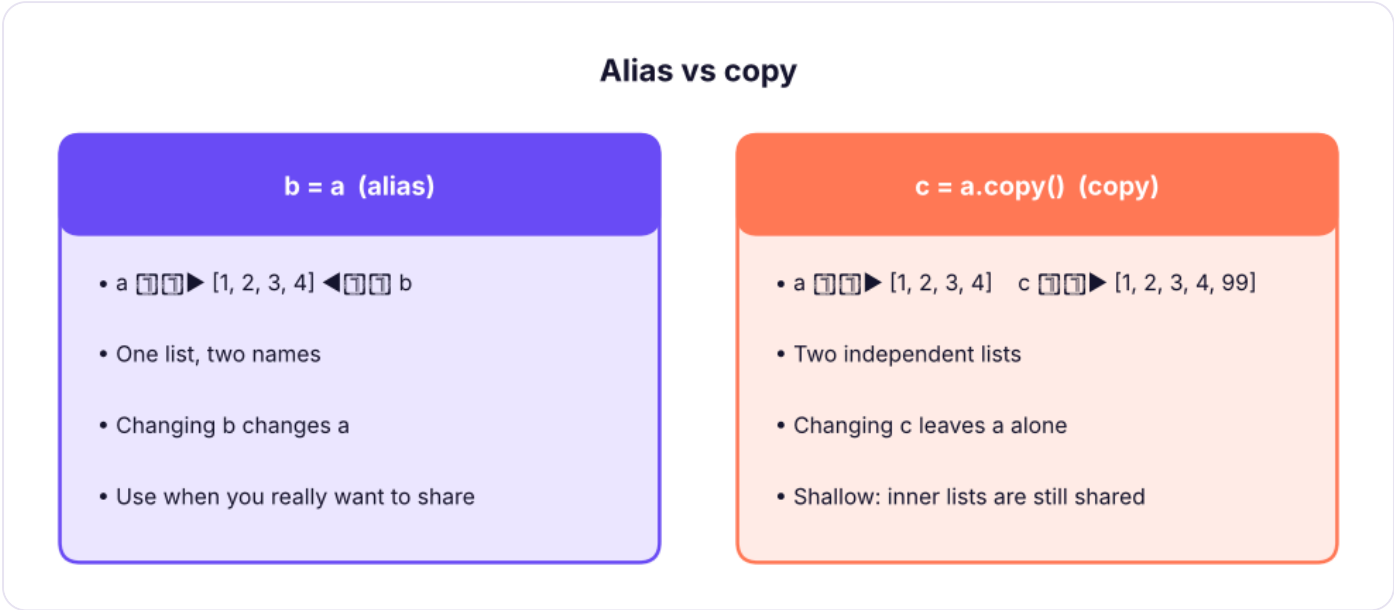
A variable holds a **reference** to a list. Assigning a list to another name does **not** copy it.

Python

```python
a = [1, 2, 3]
b = a                    # b is another name for the SAME list
b.append(4)
print(a, b, a is b)
c = a.copy()             # a real (shallow) copy; also list(a) or a[:]
c.append(99)
print(a, c, a is c)
```

OUTPUT

```
[1, 2, 3, 4] [1, 2, 3, 4] True
[1, 2, 3, 4] [1, 2, 3, 4, 99] False
```



Assignment gives two names for one list; copy() makes a second list. A shallow copy still shares inner lists.

Python

```python
import copy
grid = [[1, 2], [3, 4]]
shallow = grid.copy(); deep = copy.deepcopy(grid)
grid[0][0] = 100
print(shallow, deep)       # shallow copy sees the change in the inner list; deep copy does not
```

OUTPUT

```
[[100, 2], [3, 4]] [[1, 2], [3, 4]]
```

WATCH OUT **Mutable default arguments and `[[0] * 3] * 3`**

`[[0] * 3] * 3` creates **three references to the same inner list**; changing one row changes all. Build grids with a comprehension: `[[0] * 3 for _ in range(3)]`. Never use a list as a default function argument (`def f(x, items=[])`): use `None` and create the list inside (Lesson 9).

Nested lists

Python

```python
matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
print(matrix[1][2], [row[0] for row in matrix], [sum(row) for row in matrix])
print([list(col) for col in zip(*matrix)])                # transpose
flat = [x for row in matrix for x in row]; print(flat)    # flatten
```

OUTPUT

```
6 [1, 4, 7] [6, 15, 24]
[[1, 4, 7], [2, 5, 8], [3, 6, 9]]
[1, 2, 3, 4, 5, 6, 7, 8, 9]
```

List comprehensions

A comprehension builds a list in one readable line: `[expression for item in iterable if condition]`.

Python

```python
squares = [n ** 2 for n in range(1, 8)]
evens = [n for n in range(20) if n % 2 == 0]
upper = [c.upper() for c in ["pune", "delhi"]]
labels = ["High" if s >= 80 else "Low" for s in [95, 60, 82]]
pairs = [(i, j) for i in range(1, 4) for j in range(i, 4)]
print(squares, evens, upper, labels, pairs, sep="\n")
```

OUTPUT

```
[1, 4, 9, 16, 25, 36, 49]
[0, 2, 4, 6, 8, 10, 12, 14, 16, 18]
['PUNE', 'DELHI']
['High', 'Low', 'High']
[(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)]
```

Compare with the loop version:

Python

```python
result = []
for n in range(20):
    if n % 2 == 0:
        result.append(n)
print(result == [n for n in range(20) if n % 2 == 0])
```

OUTPUT

```
True
```

Use comprehensions for **simple** transformations and filters; switch to a loop when the logic needs several statements. For very large data use generators (Lesson 10) or pandas.

Data tasks with lists

Python

```
import pandas as pd
emp = pd.read_csv("employees_small.csv")
salaries = emp.salary.tolist(); names = emp.name.tolist(); depts = emp.department.tolist()
top3 = sorted(zip(salaries, names), reverse=True)[:3]
print("Top 3:", [(n, s) for s, n in top3])
fin = [s for s, d in zip(salaries, depts) if d == "Finance"]
print("Finance:", len(fin), "people, average", round(sum(fin) / len(fin)))
above_avg = [n for n, s in zip(names, salaries) if s > sum(salaries) / len(salaries)]
print(len(above_avg), "people earn above the average;", above_avg[:3], "…")
```

OUTPUT

```
Top 3: [('Vivaan Gupta', 157000), ('Imran Reddy', 157000), ('Krishna Khan', 156000)]
Finance: 7 people, average 95714
9 people earn above the average; ['Imran Reddy', 'Harpreet Kulkarni', 'Simran Iyer']
…
```

Python

```
# Remove duplicates but keep the original order (set() would lose the order)
raw = ["Pune", "Delhi", "Pune", "Kochi", "Delhi", "Goa"]
seen, unique = set(), []
for c in raw:
    if c not in seen: seen.add(c); unique.append(c)
print(unique, "| shortcut:", list(dict.fromkeys(raw)))
# Chunk a list into groups of 4
ids = list(range(1, 11)); print([ids[i:i + 4] for i in range(0, len(ids), 4)])
# Moving average of 3
x = [10, 12, 9, 14, 15, 13]; print([round(sum(x[i:i + 3]) / 3, 1) for i in range(len(x) -
2)])
```

OUTPUT

```
['Pune', 'Delhi', 'Kochi', 'Goa'] | shortcut: ['Pune', 'Delhi', 'Kochi', 'Goa']
[[1, 2, 3, 4], [5, 6, 7, 8], [9, 10]]
[10.3, 11.7, 12.7, 14.0]
```

Performance notes

Operation	Cost
`a[i]`, `a.append(x)`, `a.pop()`	O(1)
`x in a`, `a.index(x)`, `a.remove(x)`, `a.insert(0, x)`	O(n)
`a.sort()`	O(n log n)

For many membership tests use a **set** (O(1)); for a queue use `collections.deque`. (More in the DSA lessons.)

Summary

KEY TAKEAWAYS

- A list is an ordered, **mutable** collection; index and slice like strings; negative indexes count from the end.
- `append`, `extend`, `insert`, `remove`, `pop`, `sort`, `reverse` change the list in place and return `None`; `sorted()` and `reversed()` give new objects.
- `b = a` is an **alias**; use `a.copy()` (shallow) or `copy.deepcopy(a)` for independent copies; build grids with comprehensions.
- Sort records with `key=` (and tuple keys); sorting is stable.
- Comprehensions `[f(x) for x in xs if cond]` create lists concisely; unpack with `a, *rest = items`.
- Membership tests on lists are O(n): use sets for lookups.

Quick quiz

6 questions · tap an answer to check it

1. What does `[1, 2, 3].append(4)` return?

A [1, 2, 3, 4]

B None (the list is changed in place)

C 4

D A new list

Answer: B. Mutating methods return None.

2. After `b = a; b.append(1)`, what is true?

A a is unchanged

B a and b are the same list, so a also contains the new item

C b is a copy

D An error occurs

Answer: B. Assignment creates an alias, not a copy.

3. What does `[x * 2 for x in range(5) if x % 2 == 0]` return?

A [0, 2, 4, 6, 8]

B [0, 4, 8]

C [0, 2, 4]

D [2, 6]

Answer: B. Only x = 0, 2, 4 pass the filter and are doubled.

4. How do you sort a list of tuples by the second element, descending?

A `sorted(t)`

B `sorted(t, key=lambda p: p[1], reverse=True)`

C `t.sort(1)`

D `sort(t, 1)`

Answer: B. The key function selects the field to compare.

5. Why does `grid = [[0] * 3] * 3; grid[0][0] = 1` change every row?

A A bug in Python

B The outer multiplication repeats references to the same inner list

C Lists are immutable

D Comprehensions are required for all lists

Answer: B. Use `[[0] * 3 for _ in range(3)]`.

6. Which is the fastest way to test if an item is in a large collection many times?

A A list

B A set

C A string

D A tuple

Answer: B. Sets use hashing for O(1) average lookups.

EXERCISE

List drills

(1) From `[3, 8, 1, 9, 4, 8]` return the second-largest distinct value. (2) Remove duplicates from `["b","a","b","c","a"]` keeping order. (3) Flatten `[[1,2],[3],[4,5,6]]`. (4) Given names and salaries lists, print "name: salary" sorted by salary descending. (5) Rotate `[1,2,3,4,5]` by 2 positions to the right.

Show solution (worked answer)

Python

```python
print(sorted(set([3, 8, 1, 9, 4, 8]))[-2])                          # 8
print(list(dict.fromkeys(["b", "a", "b", "c", "a"])))               # ['b', 'a', 'c']
print([x for sub in [[1, 2], [3], [4, 5, 6]] for x in sub])         # [1, 2, 3, 4, 5, 6]
names, sal = ["A", "B", "C"], [50, 90, 70]
for n, s in sorted(zip(names, sal), key=lambda p: -p[1]): print(f"{n}: {s}")
r = [1, 2, 3, 4, 5]; print(r[-2:] + r[:-2])                         # [4, 5, 1, 2, 3]
```

INTERVIEW CORNER

Questions you may be asked

Q: Difference between a list and a tuple? Lists are mutable, tuples immutable (hashable if their items are); tuples are used for fixed records and as dictionary keys, lists for collections that change.

Q: Shallow vs deep copy? A shallow copy duplicates the outer list but shares inner objects; a deep copy recursively copies everything.

Q: What is the time complexity of `x in list` vs `x in set`? O(n) vs O(1) average.

Q: How do you reverse a list in place? `lst.reverse()` or `lst[:] = lst[::-1]`; `lst[::-1]` creates a reversed copy.

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